LAPORAN PRAKTIKUM
JARINGAN KOMPUTER 1

UJIAN AKHIR SEMESTER

Oleh:

Muhamad Septian

2213015

PROGRAM STUDI INFORMATIKA

UNIVERSITAS KATOLIK MUSI CHARITAS

PALEMBANG

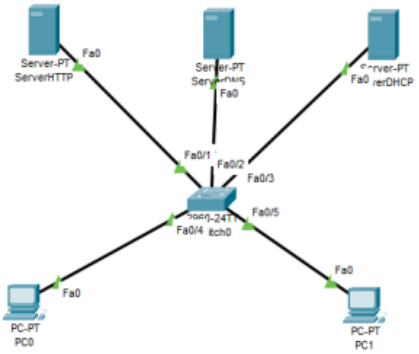
2024

Materi : Ujian Akhir Semester

Pertemuan : 13

1.

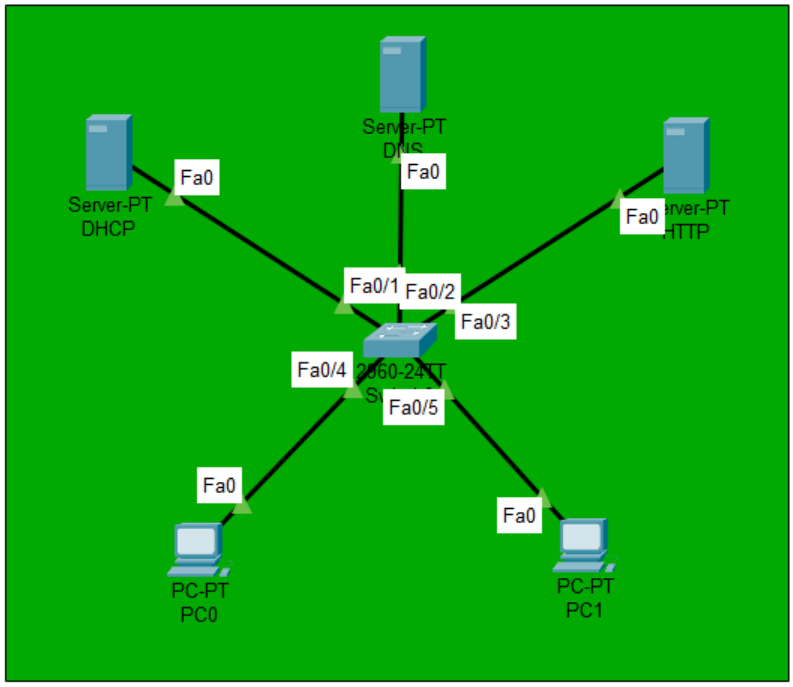
1. Lakukan konfigurasi berikut :



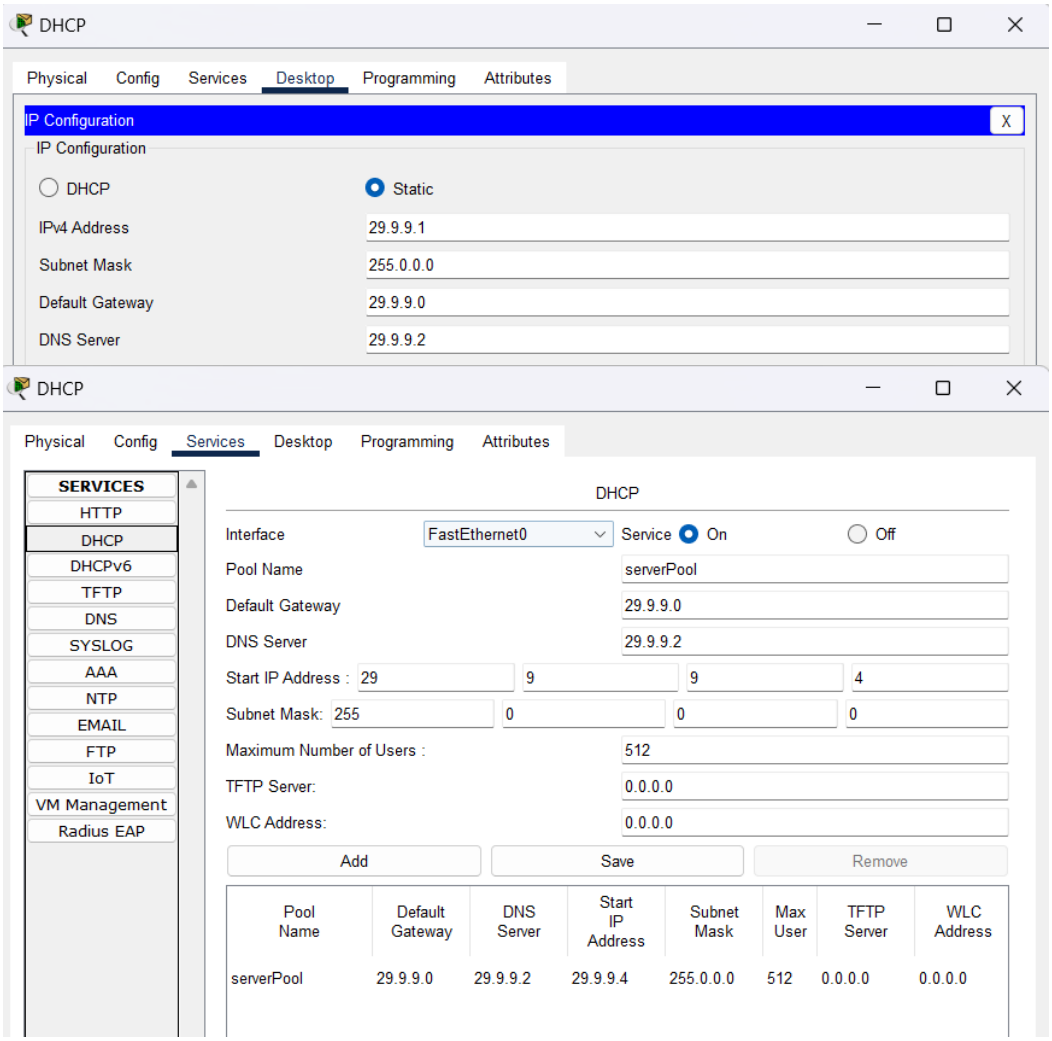
Ketentuan:

a. Terdapat tiga server, yaitu server DHCP, server DNS dan server HTTP
b. Untuk NetworkID bebas yang jelas setiap mahasiswa menggunakan network yang berbeda (poin 30)

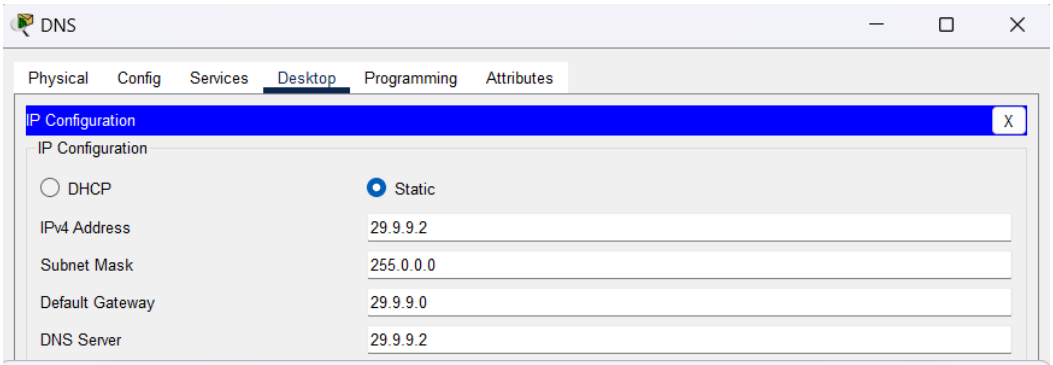
Hasil Topologi:

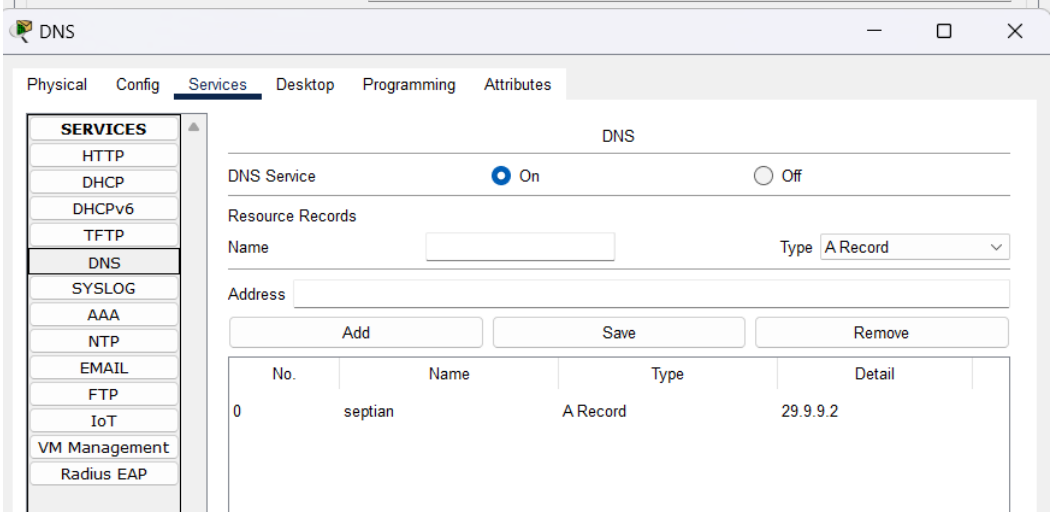


Server DHCP

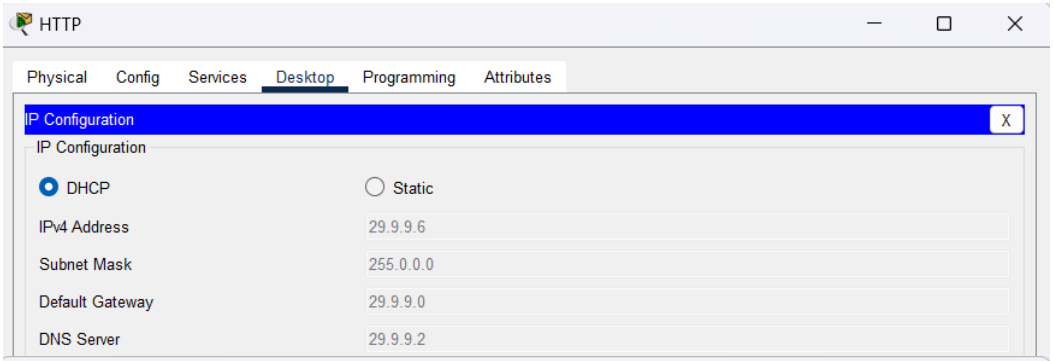

DHCP
Physical
Config
Services
Desktop
Programming
Attributes
IP Configuration
X
IP Configuration
DHCP
Static
IPv4 Address
29.9.9.1
Subnet Mask
255.0.0.0
Default Gateway
29.9.9.0
DNS Server
29.9.9.2
DHCP
Physical
Config
Services
Desktop
Programming
Attributes
SERVICES
HTTP
DHCP
DHCPv6
TFTP
DNS
SYSLOG
AAA
NTP
EMAIL
FTP
IoT
VM Management
Radius EAP
DHCP
Interface
FastEthernet0
Service
On
Off
Pool Name
serverPool
Default Gateway
29.9.9.0
DNS Server
29.9.9.2
Start IP Address :
29
9
9
4
Subnet Mask:
255
0
0
0
Maximum Number of Users :
512
TFTP Server:
0.0.0.0
WLC Address:
0.0.0.0
Add
Save
Remove
Pool Name
Default Gateway
DNS Server
Start IP Address
Subnet Mask
Max User
TFTP Server
WLC Address
serverPool
29.9.9.0
29.9.9.2
29.9.9.4
255.0.0.0
512
0.0.0.0
0.0.0.0

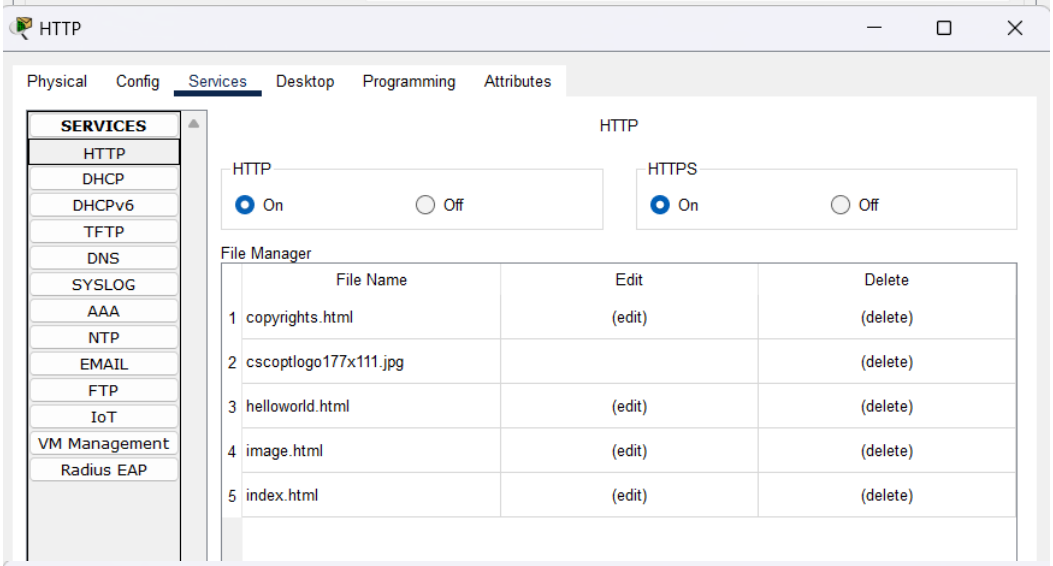
Server DNS

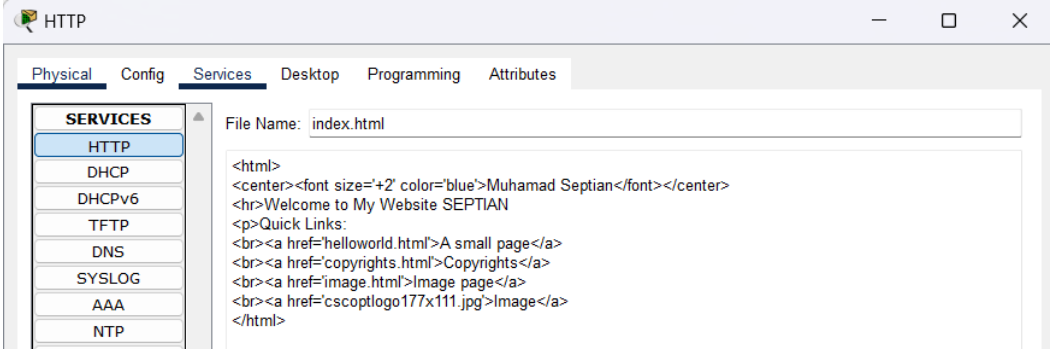

DNS
Physical
Config
Services
Desktop
Programming
Attributes
IP Configuration
X
IP Configuration
DHCP
Static
IPv4 Address
29.9.9.2
Subnet Mask
255.0.0.0
Default Gateway
29.9.9.0
DNS Server
29.9.9.2


DNS
Physical
Config
Services
Desktop
Programming
Attributes
SERVICES
HTTP
DHCP
DHCPv6
TFTP
DNS
SYSLOG
AAA
NTP
EMAIL
FTP
IoT
VM Management
Radius EAP
DNS
DNS Service
On
Off
Resource Records
Name
Type
A Record
Address
Add
Save
Remove
No.
Name
Type
Detail
0
septian
A Record
29.9.9.2

Server HTTP


HTTP
Physical
Config
Services
Desktop
Programming
Attributes
IP Configuration
X
IP Configuration
DHCP
Static
IPv4 Address
29.9.9.6
Subnet Mask
255.0.0.0
Default Gateway
29.9.9.0
DNS Server
29.9.9.2


HTTP
Physical
Config
Services
Desktop
Programming
Attributes
SERVICES
HTTP
DHCP
DHCPv6
TFTP
DNS
SYSLOG
AAA
NTP
EMAIL
FTP
IoT
VM Management
Radius EAP
HTTP
HTTP
On
Off
HTTPS
On
Off
File Manager
File Name
Edit
Delete
1 copyrights.html (edit) (delete)
2 cscoptlogo177x111.jpg (delete)
3 helloworld.html (edit) (delete)
4 image.html (edit) (delete)
5 index.html (edit) (delete)


HTTP
Physical
Config
Services
Desktop
Programming
Attributes
SERVICES
HTTP
DHCP
DHCPv6
TFTP
DNS
SYSLOG
AAA
NTP
File Name: index.html
<html>
Muhamad Septian
<hr>Welcome to My Website SEPTIAN
<p>Quick Links:

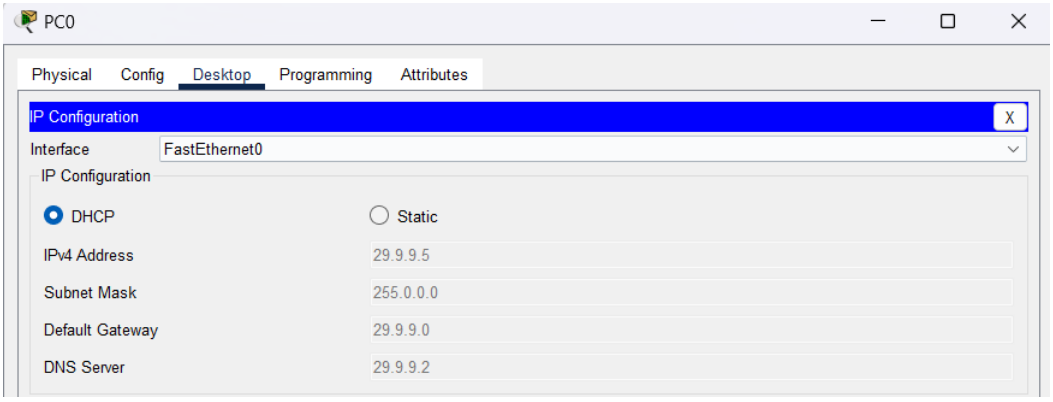
A small page

Copyrights

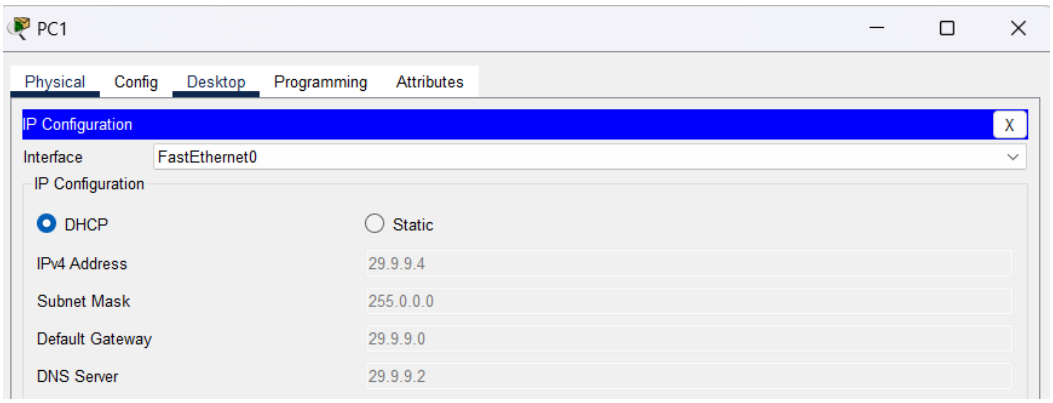
Image page

Image
</html>

PC0



PC1



Hasil Percobaan

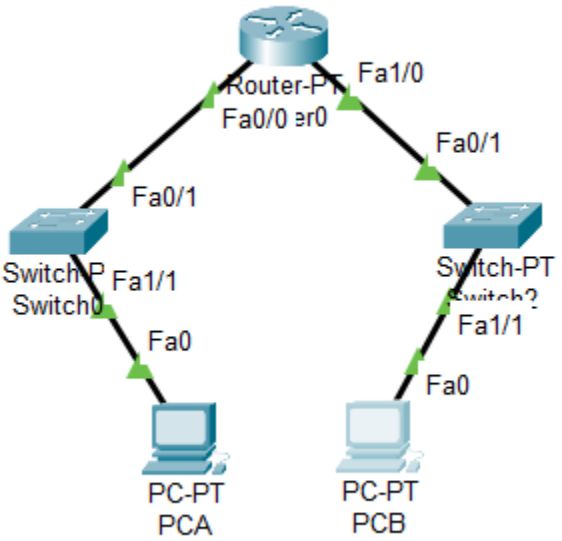
Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC0	DHCP	ICMP		0.000	N	0	(edit)	(delete)
	Successful	PC1	DNS	ICMP		0.000	N	1	(edit)	(delete)
	Successful	PC0	PC1	ICMP		0.000	N	2	(edit)	(delete)



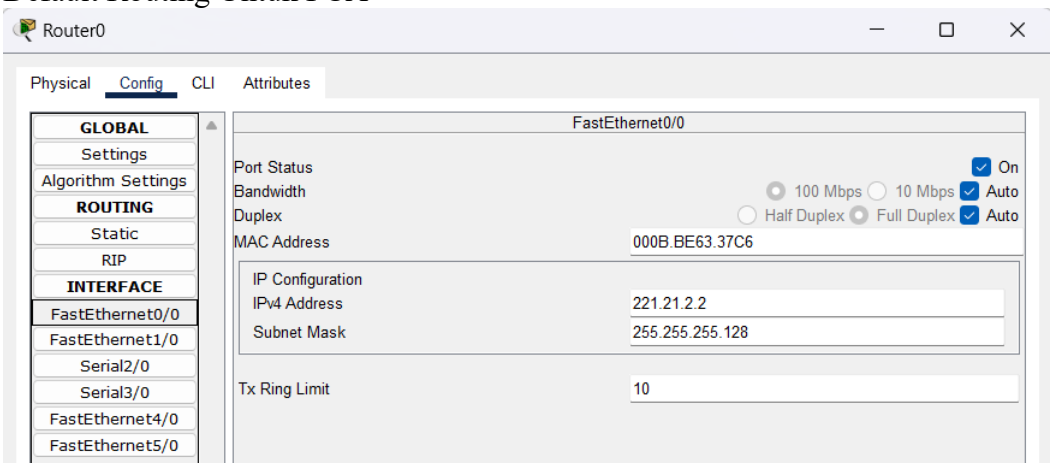
2.

2. Dalam suatu jaringan LAN terdapat dua PC, yaitu PCA dengan IP address: 221.21.2.1/25 dan PCB dengan IP adderss 150.15.1.1/16, kedua PC tersebut akan dihubungkan dengan default routing, konfigurasikan default routing tersebut (poin 30)

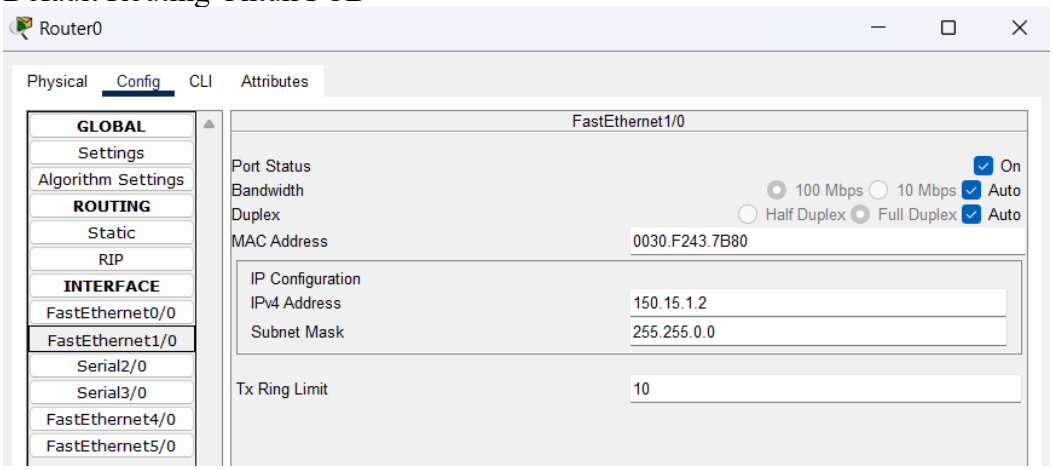
Hasil Topologi



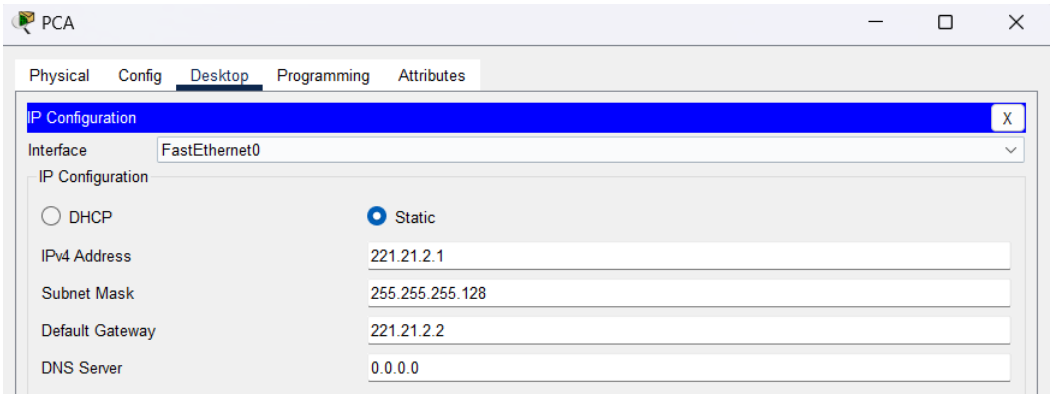
Default Routing Untuk PCA



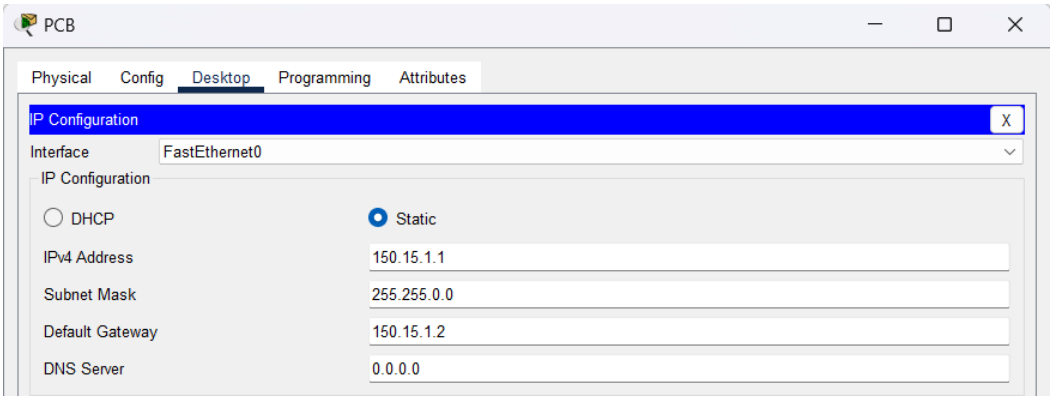
Default Routing Untuk PCB



PCA



PCB



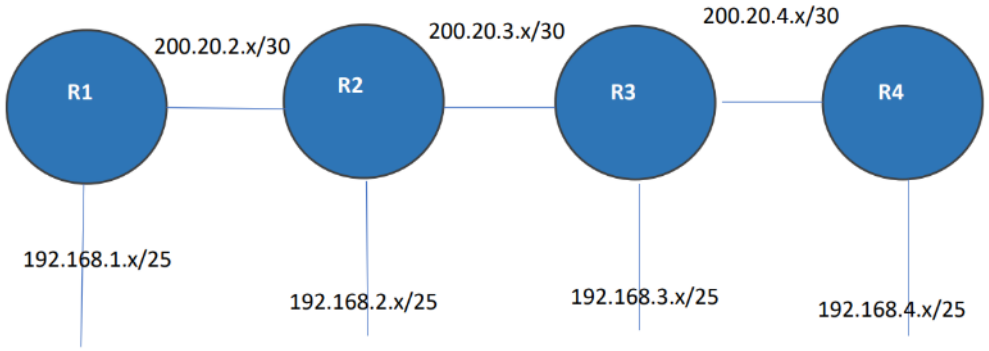
Hasil Percobaan

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PCA	Router0	ICMP		0.000	N	0	(edit)	(delete)
	Successful	PCB	Router0	ICMP		0.000	N	1	(edit)	(delete)
	Successful	PCA	PCB	ICMP		0.000	N	2	(edit)	(delete)

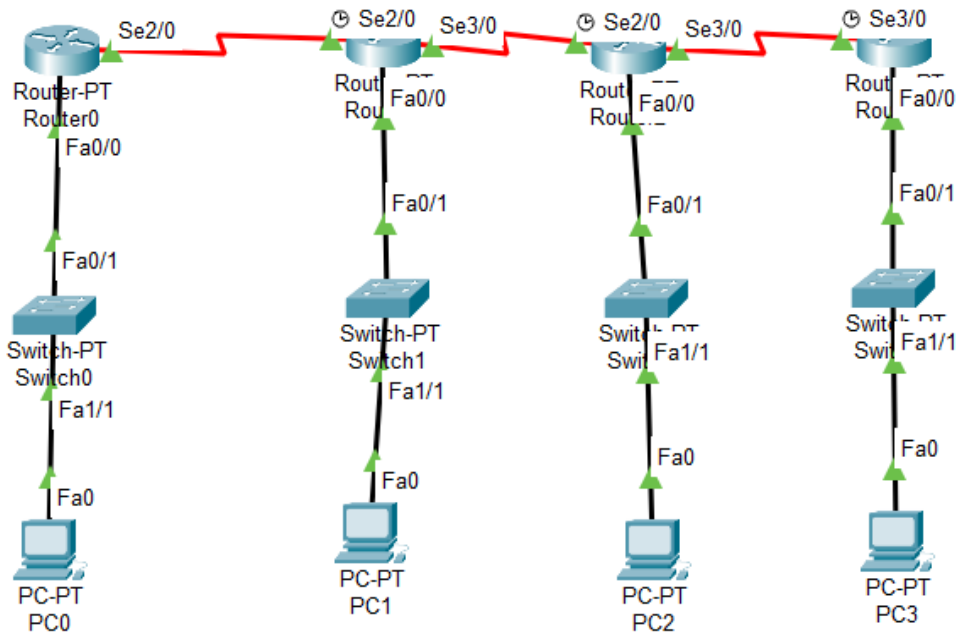
3. S

3. Konfigurasikan Routing statis pada soal teori nomor 5 (poin 40)

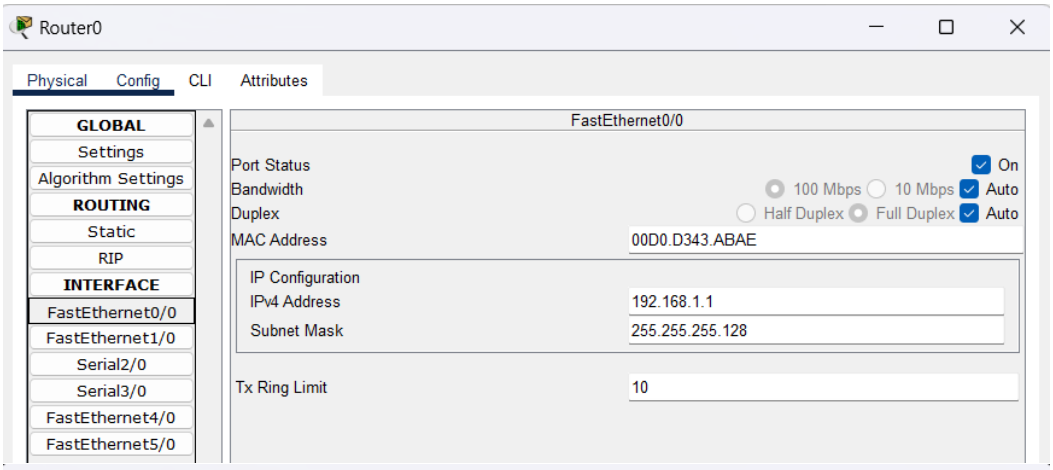
4. Terdapat topologi jaringan sebagai berikut:

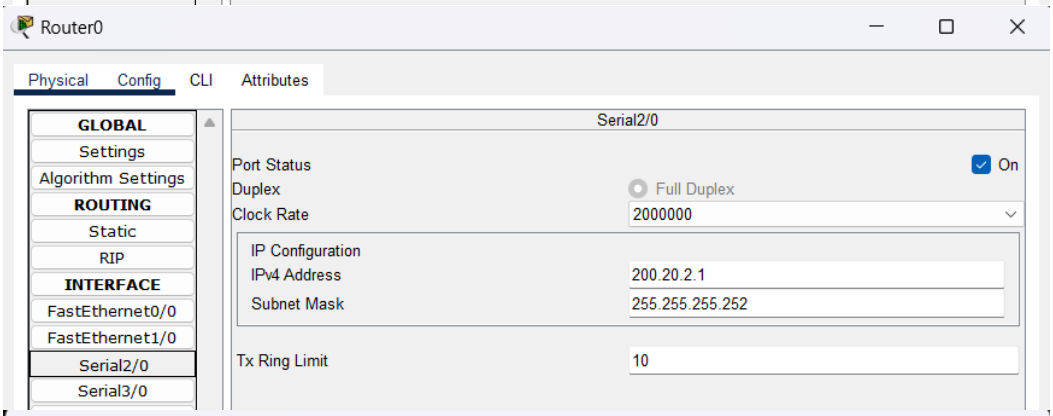


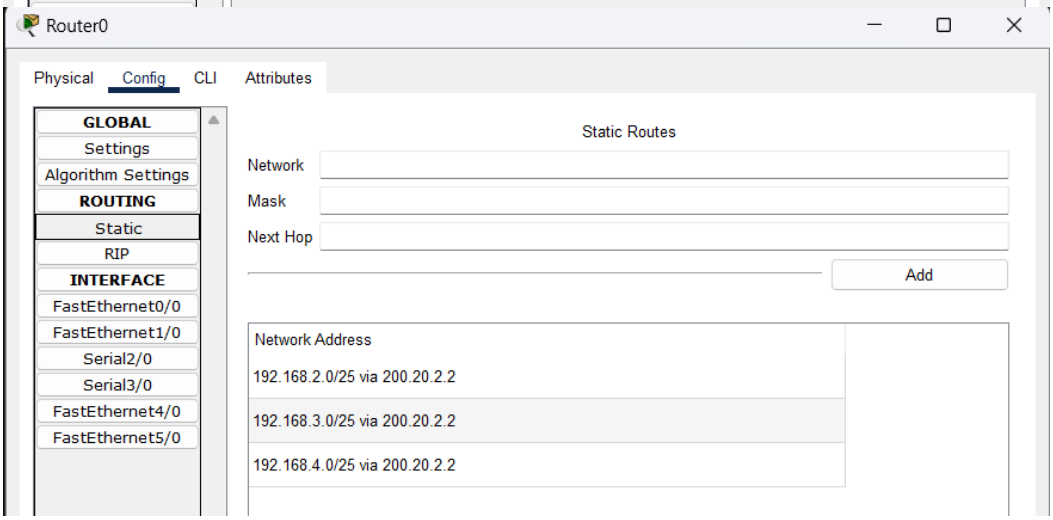
Hasil Topologi



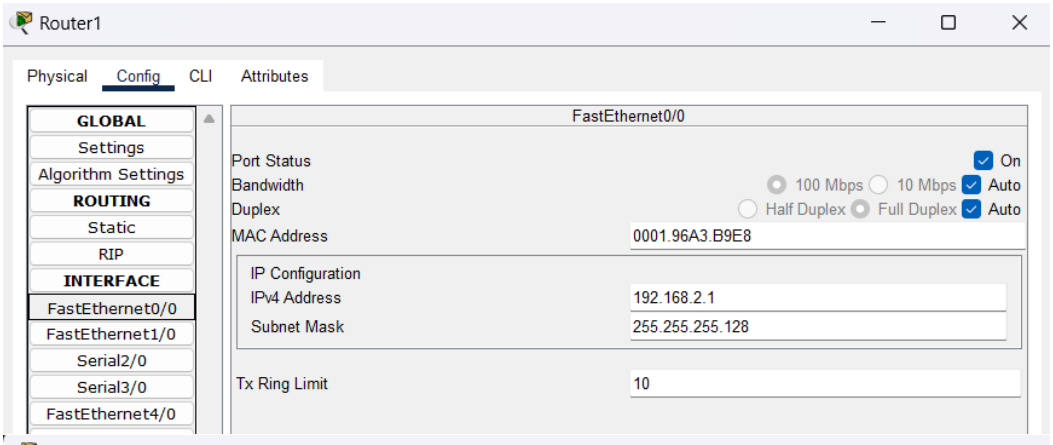
Router0

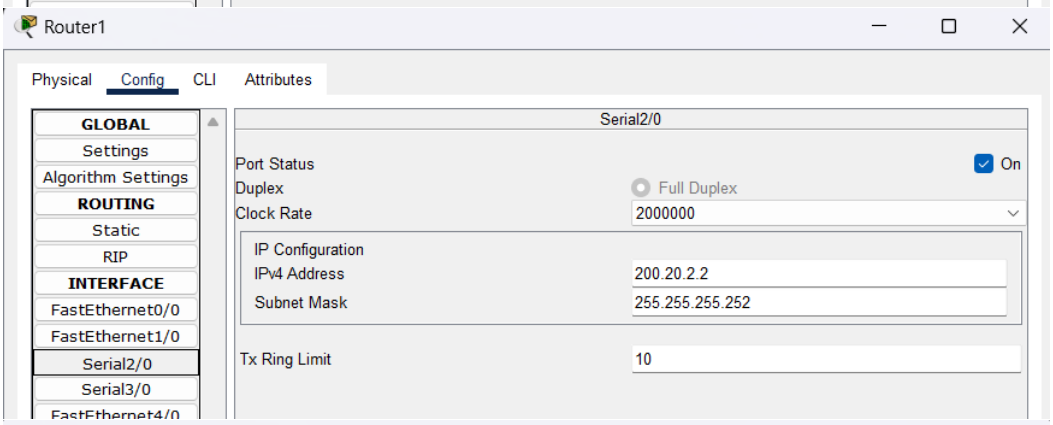

Router0
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
INTERFACE
FastEthernet0/0
FastEthernet1/0
Serial2/0
Serial3/0
FastEthernet4/0
FastEthernet5/0
FastEthernet0/0
Port Status
On
Bandwidth
100 Mbps
10 Mbps
Auto
Duplex
Half Duplex
Full Duplex
Auto
MAC Address
00D0.D343.ABAE
IP Configuration
IPv4 Address
192.168.1.1
Subnet Mask
255.255.255.128
Tx Ring Limit
10

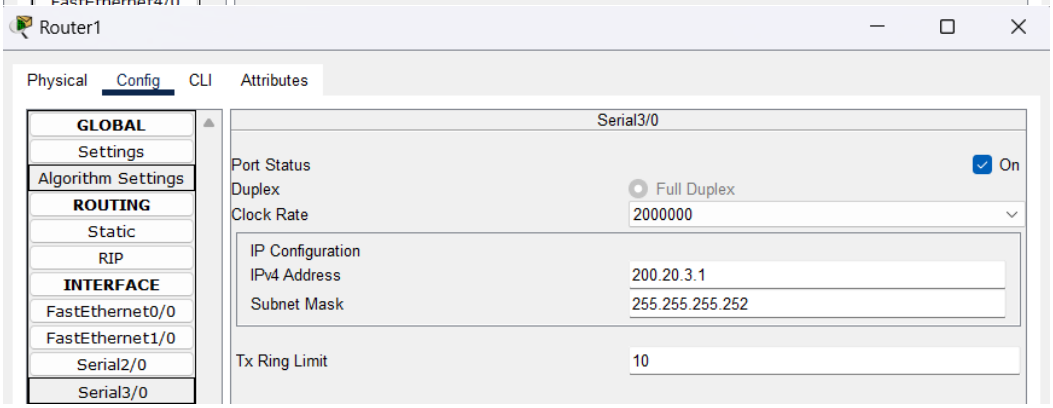

Router0
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
INTERFACE
FastEthernet0/0
FastEthernet1/0
Serial2/0
Serial3/0
Serial2/0
Port Status
On
Duplex
Full Duplex
Clock Rate
2000000
IP Configuration
IPv4 Address
200.20.2.1
Subnet Mask
255.255.255.252
Tx Ring Limit
10

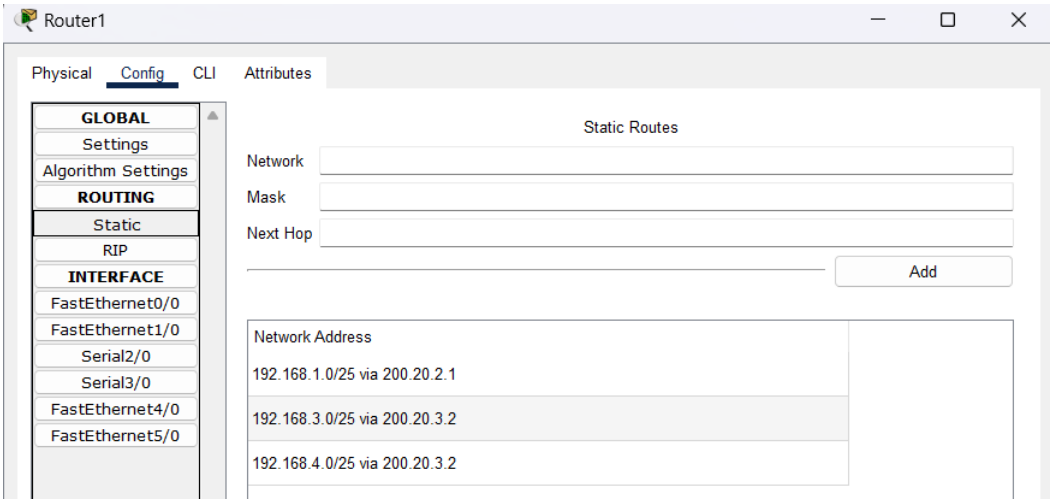

Router0
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
INTERFACE
FastEthernet0/0
FastEthernet1/0
Serial2/0
Serial3/0
FastEthernet4/0
FastEthernet5/0
Static Routes
Network
Mask
Next Hop
Add
Network Address
192.168.2.0/25 via 200.20.2.2
192.168.3.0/25 via 200.20.2.2
192.168.4.0/25 via 200.20.2.2

Router1

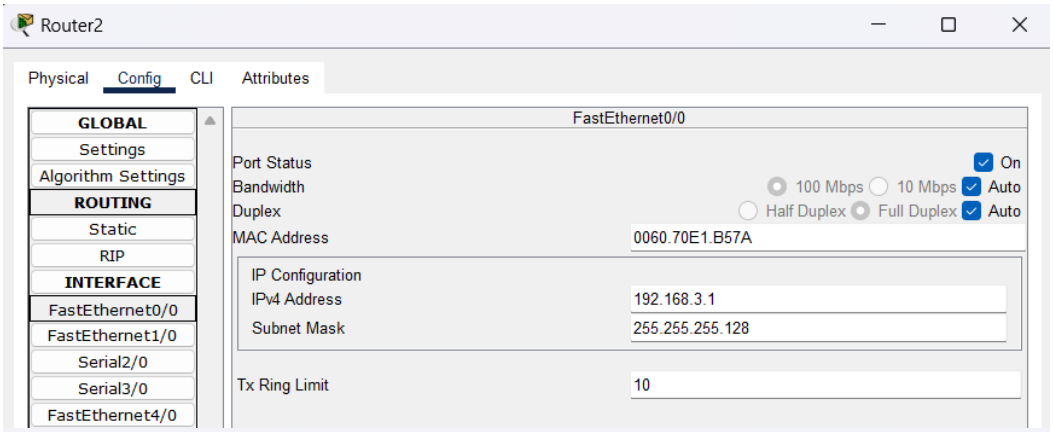

Router1
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
INTERFACE
FastEthernet0/0
FastEthernet1/0
Serial2/0
Serial3/0
FastEthernet4/0
FastEthernet0/0
Port Status
On
Bandwidth
100 Mbps
10 Mbps
Auto
Duplex
Half Duplex
Full Duplex
Auto
MAC Address
0001.96A3.B9E8
IP Configuration
IPv4 Address
192.168.2.1
Subnet Mask
255.255.255.128
Tx Ring Limit
10

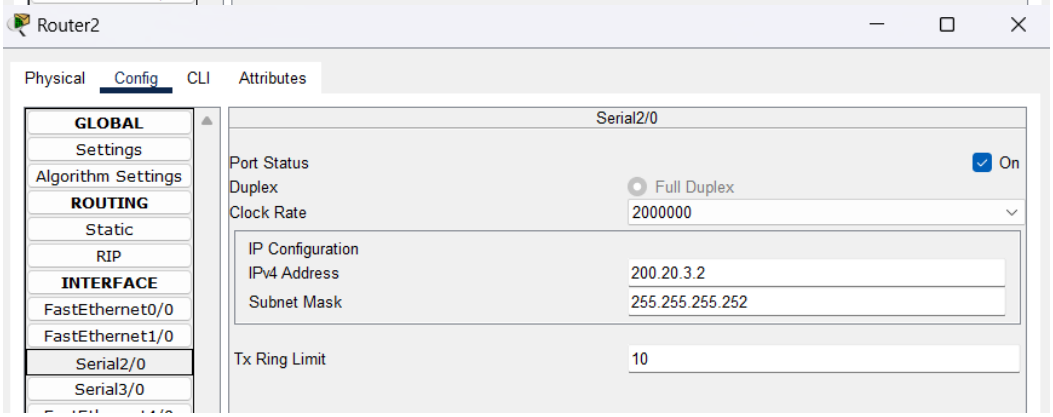

Router1
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
INTERFACE
FastEthernet0/0
FastEthernet1/0
Serial2/0
Serial3/0
Serial2/0
Port Status
On
Duplex
Full Duplex
Clock Rate
2000000
IP Configuration
IPv4 Address
200.20.2.2
Subnet Mask
255.255.255.252
Tx Ring Limit
10

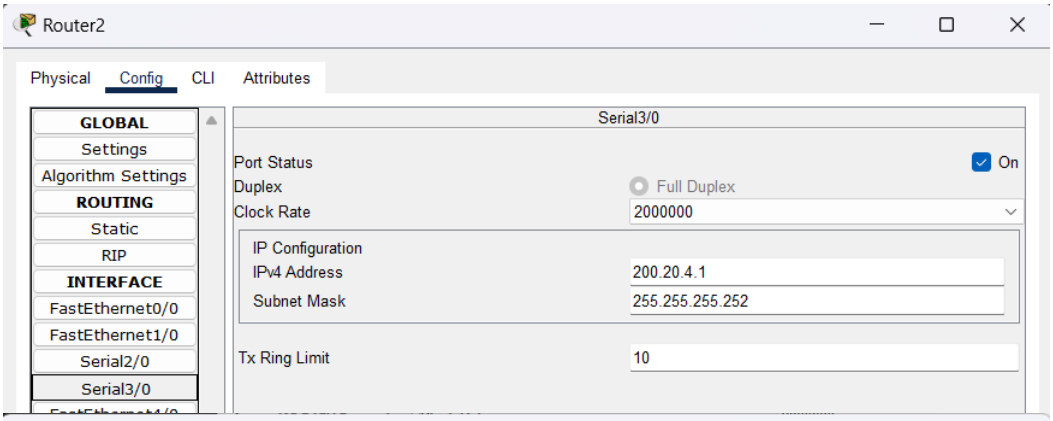

Router1
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
INTERFACE
FastEthernet0/0
FastEthernet1/0
Serial2/0
Serial3/0
Serial3/0
Port Status
On
Duplex
Full Duplex
Clock Rate
2000000
IP Configuration
IPv4 Address
200.20.3.1
Subnet Mask
255.255.255.252
Tx Ring Limit
10

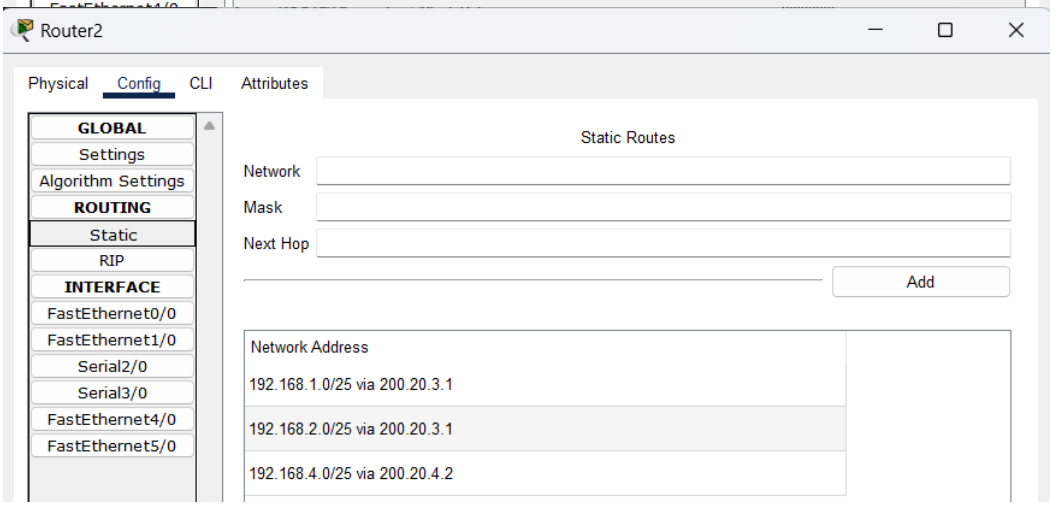

Router1
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
INTERFACE
FastEthernet0/0
FastEthernet1/0
Serial2/0
Serial3/0
FastEthernet4/0
FastEthernet5/0
Static Routes
Network
Mask
Next Hop
Add
Network Address
192.168.1.0/25 via 200.20.2.1
192.168.3.0/25 via 200.20.3.2
192.168.4.0/25 via 200.20.3.2

Router2

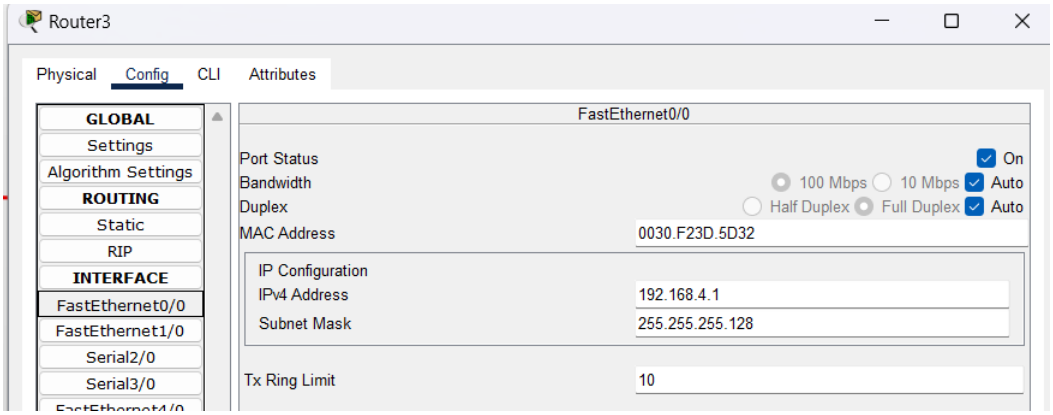

Router2
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
INTERFACE
FastEthernet0/0
FastEthernet1/0
Serial2/0
Serial3/0
FastEthernet4/0
FastEthernet0/0
Port Status
On
Bandwidth
100 Mbps
10 Mbps
Auto
Duplex
Half Duplex
Full Duplex
Auto
MAC Address
0060.70E1.B57A
IP Configuration
IPv4 Address
192.168.3.1
Subnet Mask
255.255.255.128
Tx Ring Limit
10

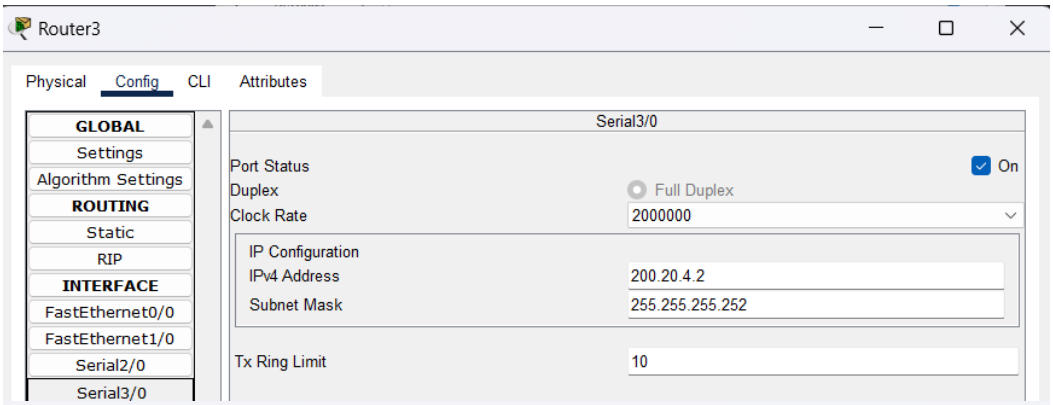

Router2
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
INTERFACE
FastEthernet0/0
FastEthernet1/0
Serial2/0
Serial3/0
Serial2/0
Port Status
On
Duplex
Full Duplex
Clock Rate
2000000
IP Configuration
IPv4 Address
200.20.3.2
Subnet Mask
255.255.255.252
Tx Ring Limit
10

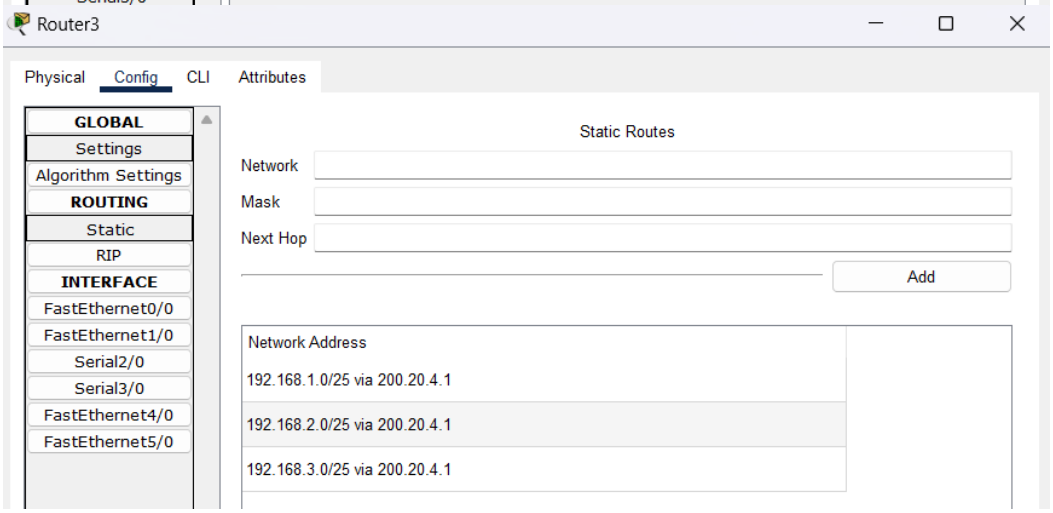

Router2
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
INTERFACE
FastEthernet0/0
FastEthernet1/0
Serial2/0
Serial3/0
Serial3/0
Port Status
On
Duplex
Full Duplex
Clock Rate
2000000
IP Configuration
IPv4 Address
200.20.4.1
Subnet Mask
255.255.255.252
Tx Ring Limit
10


Router2
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
INTERFACE
FastEthernet0/0
FastEthernet1/0
Serial2/0
Serial3/0
FastEthernet4/0
FastEthernet5/0
Static Routes
Network
Mask
Next Hop
Add
Network Address
192.168.1.0/25 via 200.20.3.1
192.168.2.0/25 via 200.20.3.1
192.168.4.0/25 via 200.20.4.2

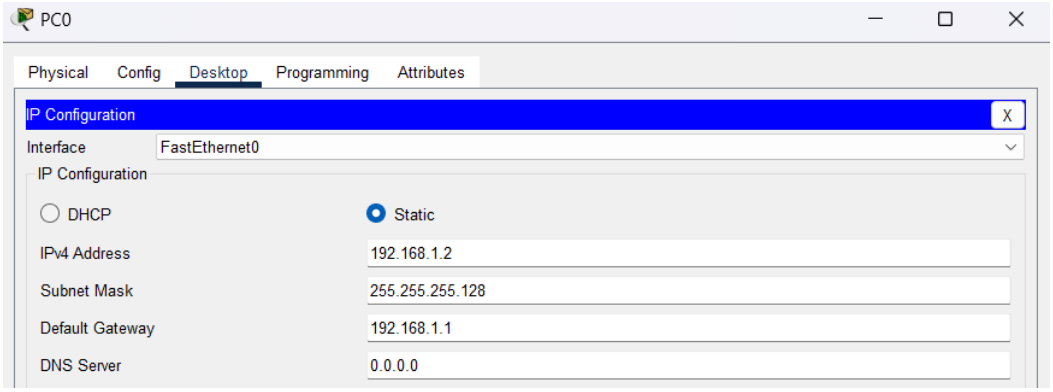
Router3


Router3
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
INTERFACE
FastEthernet0/0
FastEthernet1/0
Serial2/0
Serial3/0
FastEthernet0/0
Port Status
On
Bandwidth
100 Mbps
10 Mbps
Auto
Duplex
Half Duplex
Full Duplex
Auto
MAC Address
0030.F23D.5D32
IP Configuration
IPv4 Address
192.168.4.1
Subnet Mask
255.255.255.128
Tx Ring Limit
10

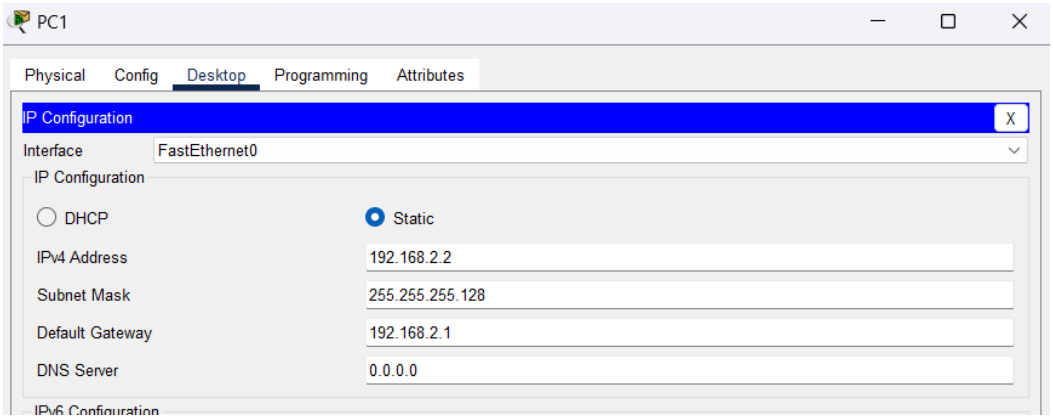

Router3
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
INTERFACE
FastEthernet0/0
FastEthernet1/0
Serial2/0
Serial3/0
Serial3/0
Port Status
On
Duplex
Full Duplex
Clock Rate
2000000
IP Configuration
IPv4 Address
200.20.4.2
Subnet Mask
255.255.255.252
Tx Ring Limit
10


Router3
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
INTERFACE
FastEthernet0/0
FastEthernet1/0
Serial2/0
Serial3/0
FastEthernet4/0
FastEthernet5/0
Static Routes
Network
Mask
Next Hop
Add
Network Address
192.168.1.0/25 via 200.20.4.1
192.168.2.0/25 via 200.20.4.1
192.168.3.0/25 via 200.20.4.1

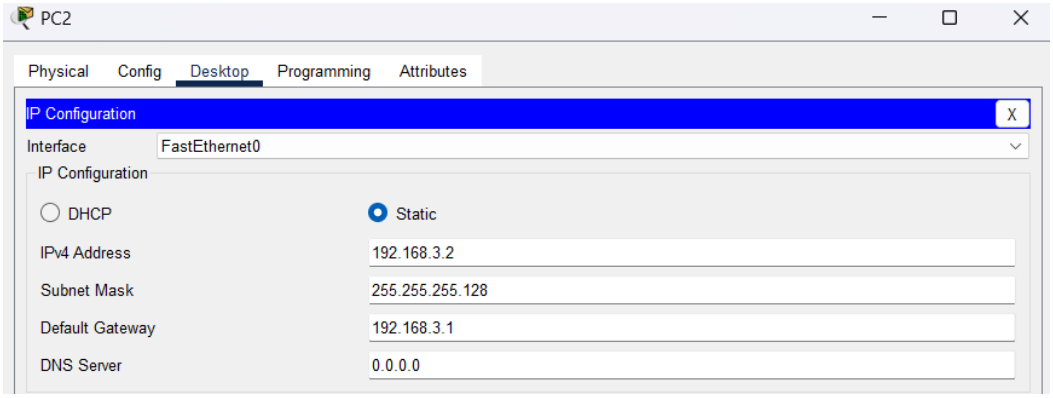
PC0


PC0
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.1.2
Subnet Mask
255.255.255.128
Default Gateway
192.168.1.1
DNS Server
0.0.0.0

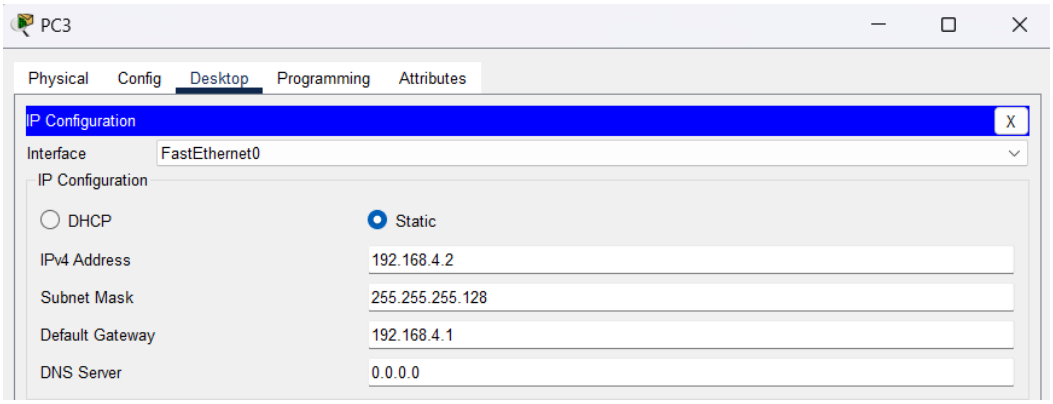
PC1


PC1
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.2.2
Subnet Mask
255.255.255.128
Default Gateway
192.168.2.1
DNS Server
0.0.0.0

PC2


PC2
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.3.2
Subnet Mask
255.255.255.128
Default Gateway
192.168.3.1
DNS Server
0.0.0.0

PC3


PC3
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.4.2
Subnet Mask
255.255.255.128
Default Gateway
192.168.4.1
DNS Server
0.0.0.0

Hasil Percobaan

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC0	Router0	ICMP		0.000	N	0	(edit)	(delete)
	Successful	PC1	Router1	ICMP		0.000	N	1	(edit)	(delete)
	Successful	PC2	Router2	ICMP		0.000	N	2	(edit)	(delete)
	Successful	PC3	Router3	ICMP		0.000	N	3	(edit)	(delete)

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC3	Router3	ICMP		0.000	N	3	(edit)	(delete)
	Successful	Router0	Router1	ICMP		0.000	N	4	(edit)	(delete)
	Successful	Router2	Router3	ICMP		0.000	N	5	(edit)	(delete)
	Successful	PC0	PC1	ICMP		0.000	N	6	(edit)	(delete)

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	Router2	Router3	ICMP		0.000	N	5	(edit)	(delete)
	Successful	PC0	PC1	ICMP		0.000	N	6	(edit)	(delete)
	Successful	PC2	PC3	ICMP		0.000	N	7	(edit)	(delete)